

```
<h1>Hello Students</h1>
```

JavaScript can select and modify it.

Dynamic HTML :

```
const heading =
```

```
document.createElement("h1");
```

Static

HTML → Browser → DOM

Dynamic
JavaScript

↓
createElement()

↓
New DOM element

↓
append()

1. createElement ()

createElement creates new HTML Elements

```
const element =  
document.createElement("tagName");
```

```
const element =  
document.createElement("tagName");
```

Creating an element does not automatically display it on the webpage.

We need to add it to the DOM.

2. Creating Element step by step

`querySelector( )` selects the first element that matches a css Selector

```
<body>
  <div id="app"></div>
</body>
const app = document.getElementById("app");
const heading = document.createElement("h1");
heading.innerText = "Welcome Students";
app.append(heading);
```

Output :

```
<div id="app">
  <h1>Welcome Students</h1>
</div>
```

The Four step pattern :

1. SELECT parent



2. CREATE element



3. MODIFY element



4. APPEND element

3. append()

Append adds content or elements at the end of the parent Element

```
const list = document.querySelector("#studentList");  
  
const student = document.createElement("li");  
  
student.innerText = "Arun";  
  
list.append(student);  
  
<ul id="studentList">  
  <li>Arun</li>  
</ul>
```

4. Adding multiple Elements

```
<ul id="studentList"></ul>
const studentList = document.querySelector("#studentList");

const student1 = document.createElement("li");
student1.innerText = "Arun";

const student2 = document.createElement("li");
student2.innerText = "Priya";

const student3 = document.createElement("li");
student3.innerText = "Kumar";

studentList.append(student1);
studentList.append(student2);
studentList.append(student3);
```

```
const students = ["Arun", "Priya", "Kumar"];

const studentList =
document.querySelector("#studentList");

students.forEach((studentName) => {
  const student = document.createElement("li");

  student.innerText = studentName;

  studentList.append(student);
});
```

5. Adding Css

```
.student {  
  padding: 10px;  
  margin: 5px;  
}  
const student = document.createElement("li");  
  
student.innerText = "Arun";  
  
student.classList.add("student");  
  
studentList.append(student);  
Result :  
<li class="student">Arun</li>
```

6. Creating Complex Elements

```
const card = document.createElement("div");

const title = document.createElement("h2");
title.innerText = "JavaScript";

const description = document.createElement("p");
description.innerText = "Learn DOM manipulation.";

card.append(title);
card.append(description);

document.body.append(card);
Output :
<div>
  <h2>JavaScript</h2>
  <p>Learn DOM manipulation.</p>
</div>
```

7. remove

Removes an element from DOM

```
const heading = document.querySelector("h1");  
heading.remove();
```

Remove a Dynamically created Element

```
const list = document.querySelector("#studentList");  
const student = document.createElement("li");  
student.innerText = "Arun";  
list.append(student);  
student.remove();
```

7. Remove specific element

Removes an element from DOM

```
<ul id="studentList">  
  <li class="student">Arun</li>  
  <li class="student">Priya</li>  
  <li class="student">Kumar</li>  
</ul>
```

```
const student =  
document.querySelector(".student");  
  
student.remove();
```

Only the first matching student is removed.

7. Remove All element

Removes an element from DOM

```
<ul id="studentList">  
  <li class="student">Arun</li>  
  <li class="student">Priya</li>  
  <li class="student">Kumar</li>  
</ul>
```

```
const students =  
document.querySelectorAll(".student");
```

```
students.forEach((student) => {  
  student.remove();  
});
```

Now all .student are removed